M54 Schematic Documentation

Project: BladeCore   Board: M54   Variant: E   Rev 1.0.0   2026-02-07

CONTENT

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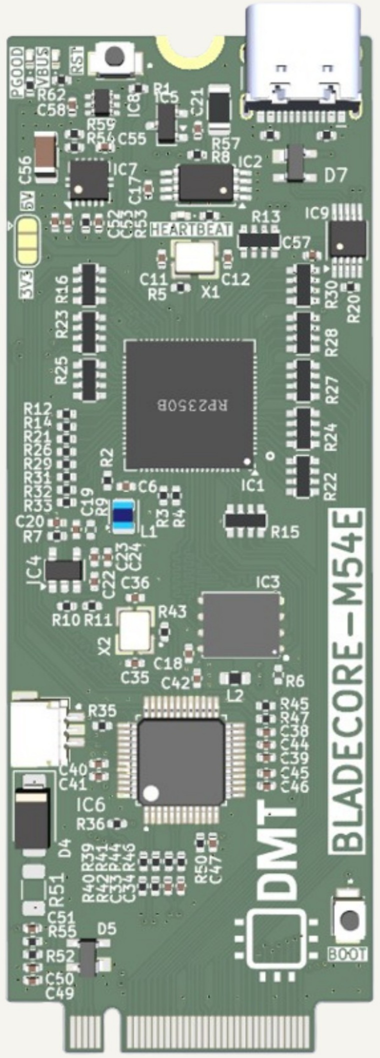
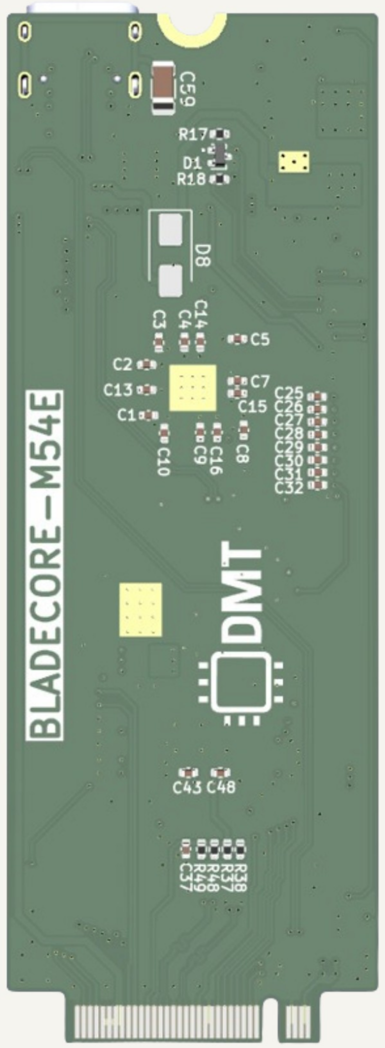
PCB PREVIEW

TOP VIEW



BOTTOM VIEW



COMMENT GUIDELINES

General comments are black, 50 mil size

Design notes and guidelines are blue, 50 mil size

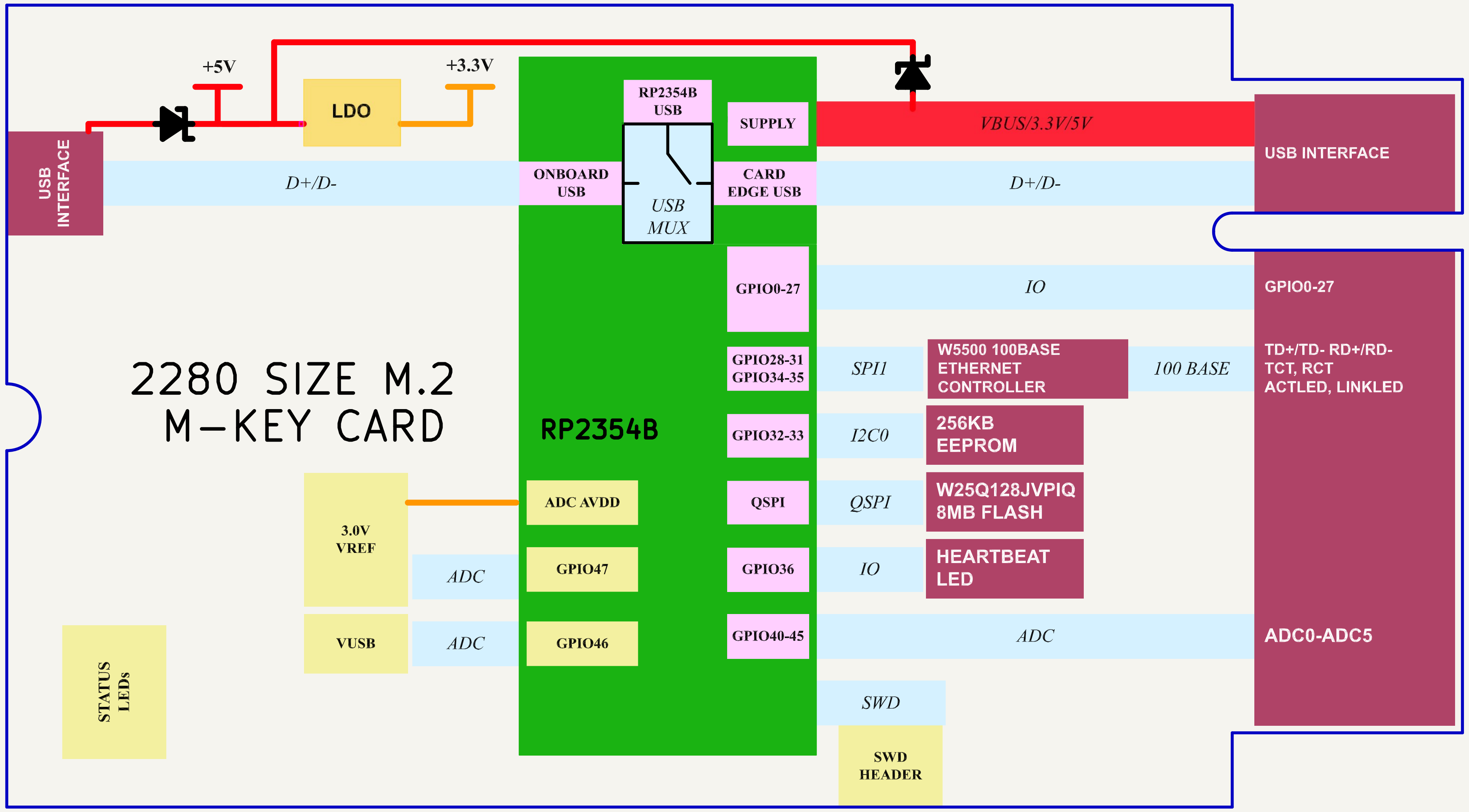
Layout instructions are red, 50 mil size

NOTES

Not fitted components are marked as X

DMT	Board Name: BladeCore	Project Name: M54	
Date: 2026-01-31	File Name: BladeCore-M54E.kicad_sch	Revision: 1.0.0	Variant: E
Sheet Title: Root	Company: DvidMakesThings; Designer: David Sipos; Reviewer:	Size: A3	Sheet: 1 of 9

[2] Block Diagram

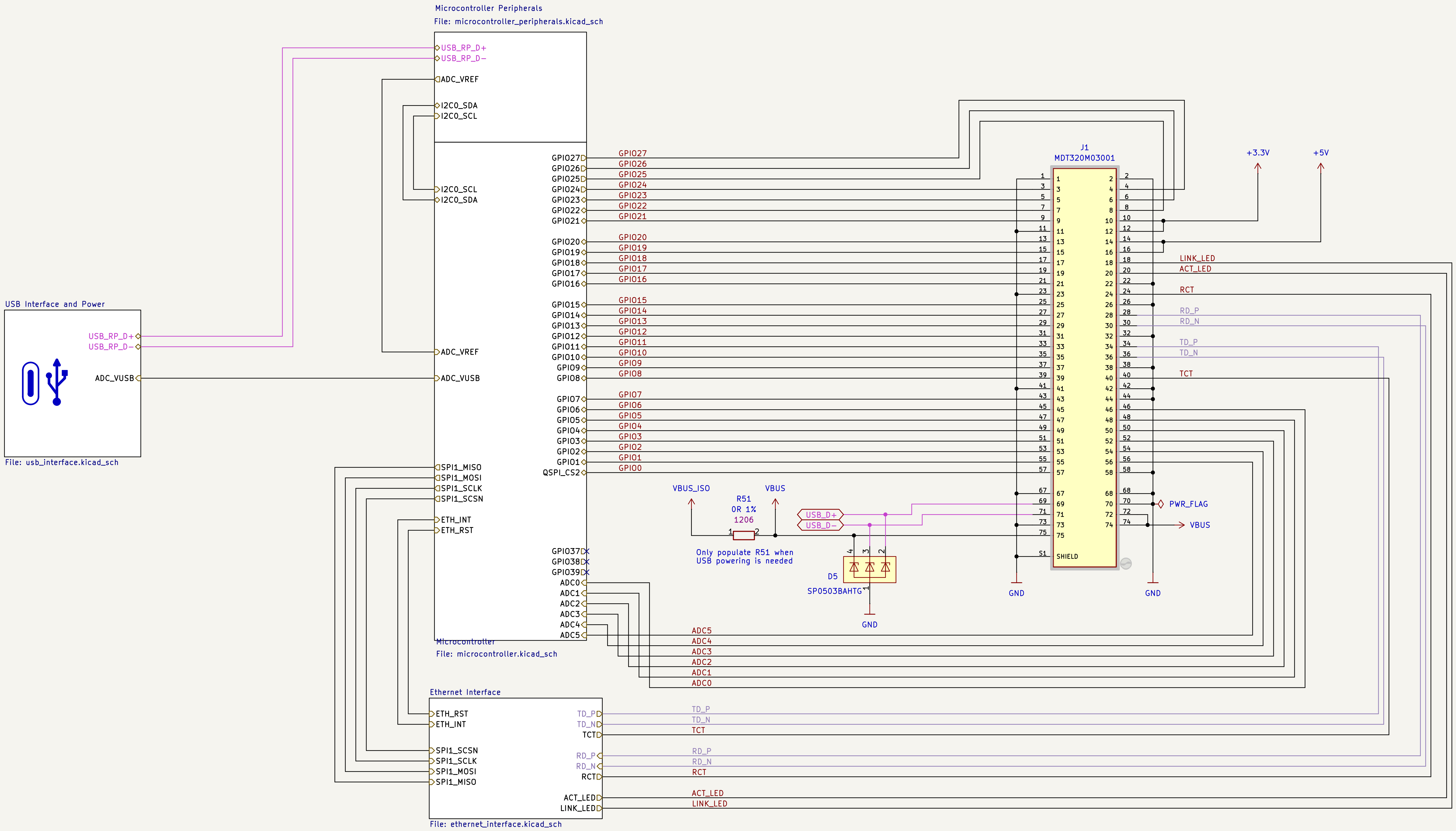


Electrical specifications:

Input voltage:	3.3V - 5.5 V
Absolut maximum current consumption	480 mA
Typical current consumtion	60mA
IO voltage	3.3V
Onboard LDO output capacity	1.5A
LDO Dropout	60mV
EEPROM	256 kbit
External Flash	8MB

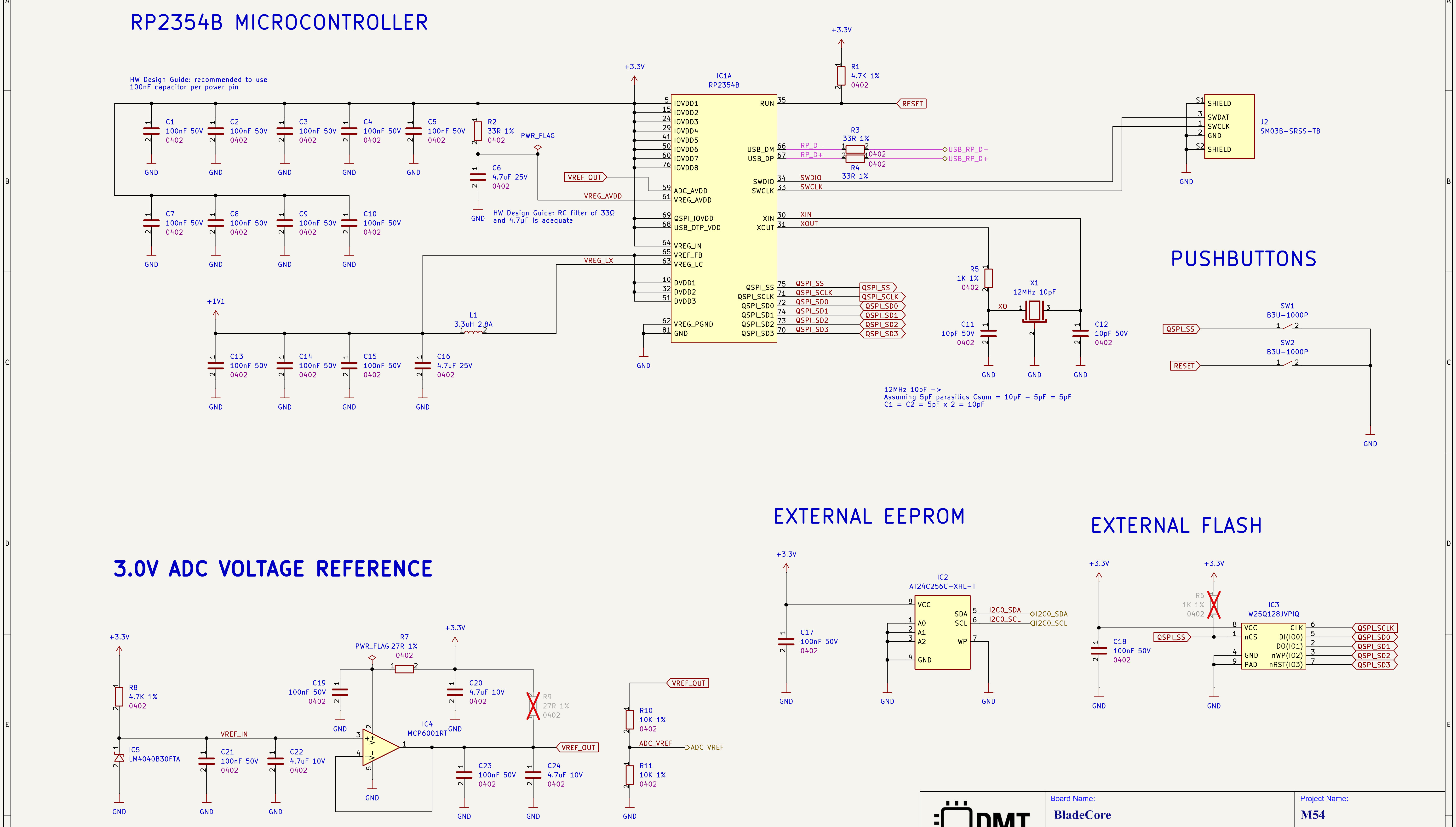
DMT Date: 2026-01-31	Board Name: BladeCore File Name: Block Diagram.kicad_sch		Project Name: M54 Revision: 1.0.0	Variant: E
Sheet Title: Block Diagram	Company: DvidMakesThings	Designer: David Sipos Reviewer:	Size: A3	Sheet: 2 of 9

[3] Project Architecture



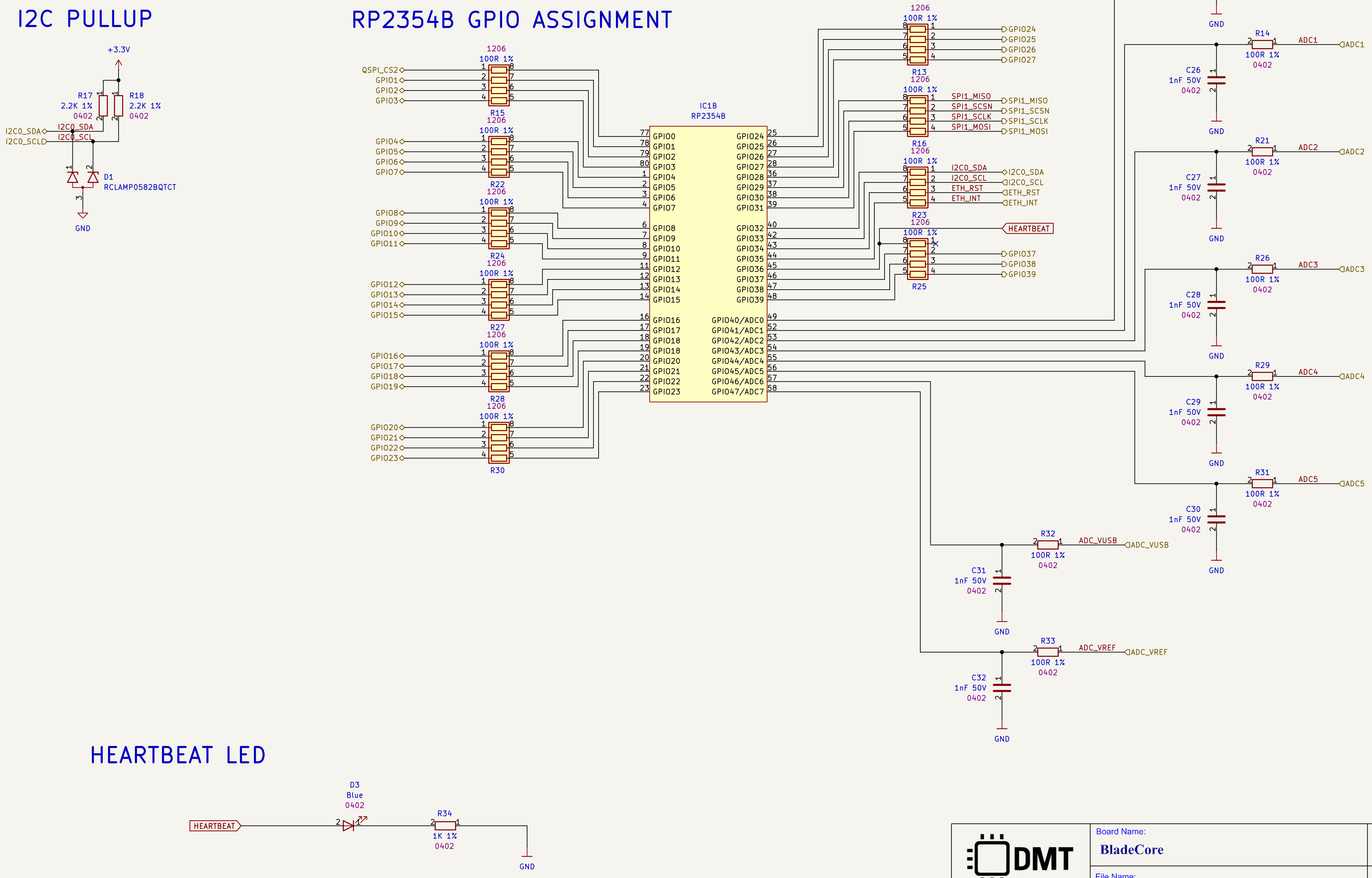
Board Name: BladeCore		Project Name: M54
File Name: Project Architecture.kicad_sch		Revision: 1.0.0 / Variant: E
Date: 2026-01-31		
Sheet Title: Project Architecture	Company: DvidMakesThings / Designer: David Sipos / Reviewer:	Size: A3 / Sheet: 3 of 9

[4] Microcontroller Peripherals



DMT	Board Name: BladeCore		Project Name: M54	
Date: 2026-01-31	File Name: microcontroller_peripherals.kicad_sch		Revision: 1.0.0	Variant: E
Sheet Title: Microcontroller Peripherals	Company: DvidMakesThings	Designer: David Sipos Reviewer:	Size: A3	Sheet: 4 of 9

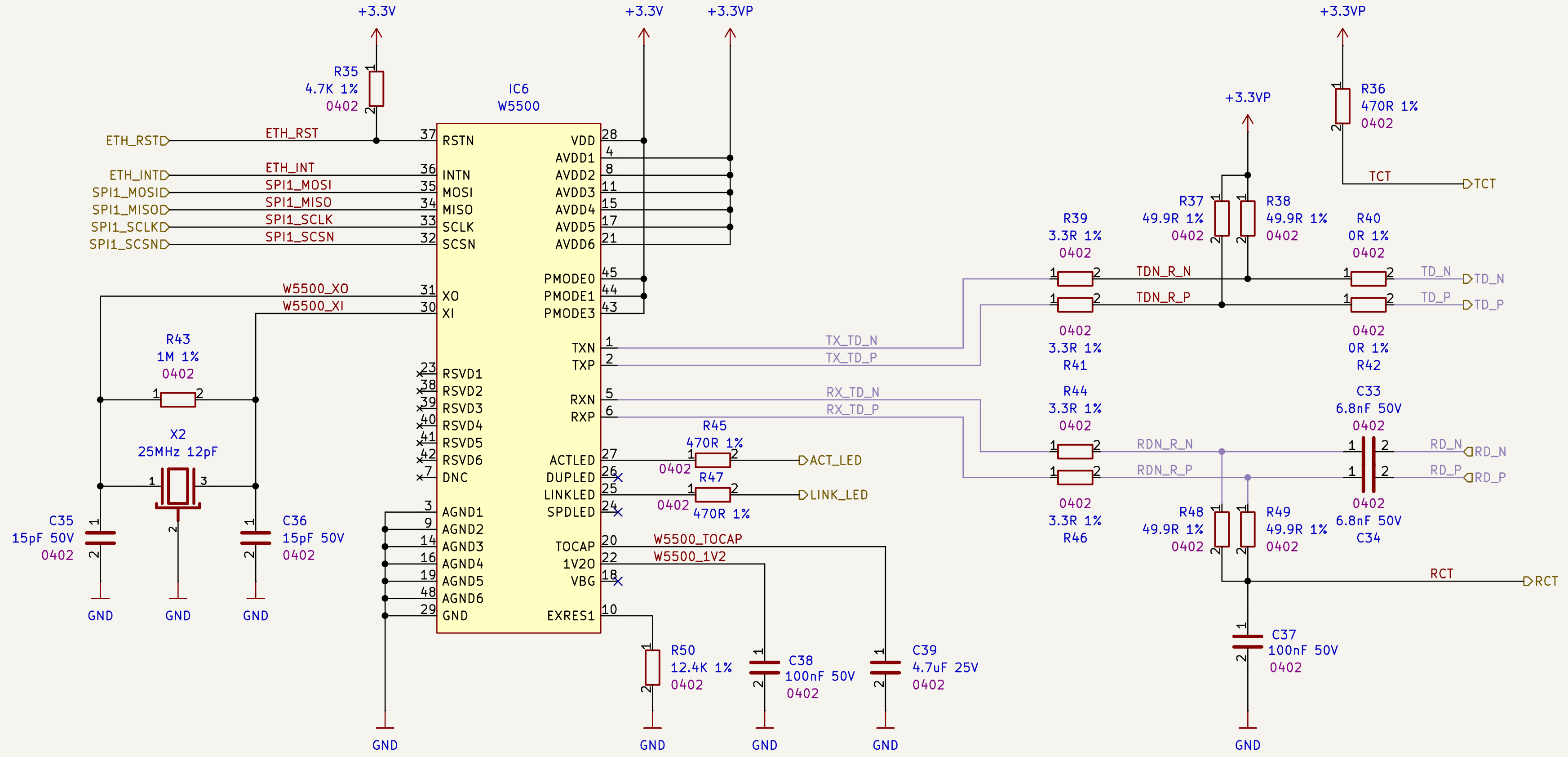
[5] Microcontroller



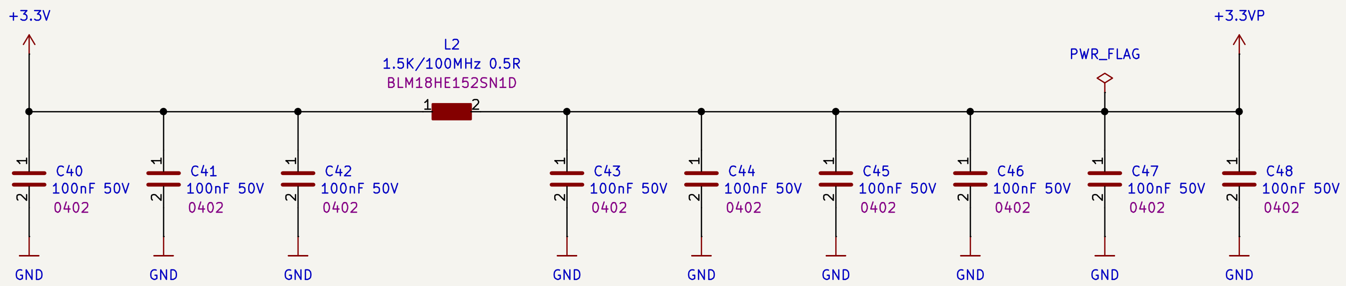
DMT	Board Name: BladeCore		Project Name: M54	
Date: 2026-01-31	File Name: microcontroller.kicad_sch		Revision: 1.0.0	Variant: E
Sheet Title: Microcontroller	Company: DvidMakesThings	Designer: David Sipos Reviewer:	Size: A3	Sheet: 5 of 9

[6] Ethernet Interface

W5500 ETHERNET CONTROLLER WITH PHY

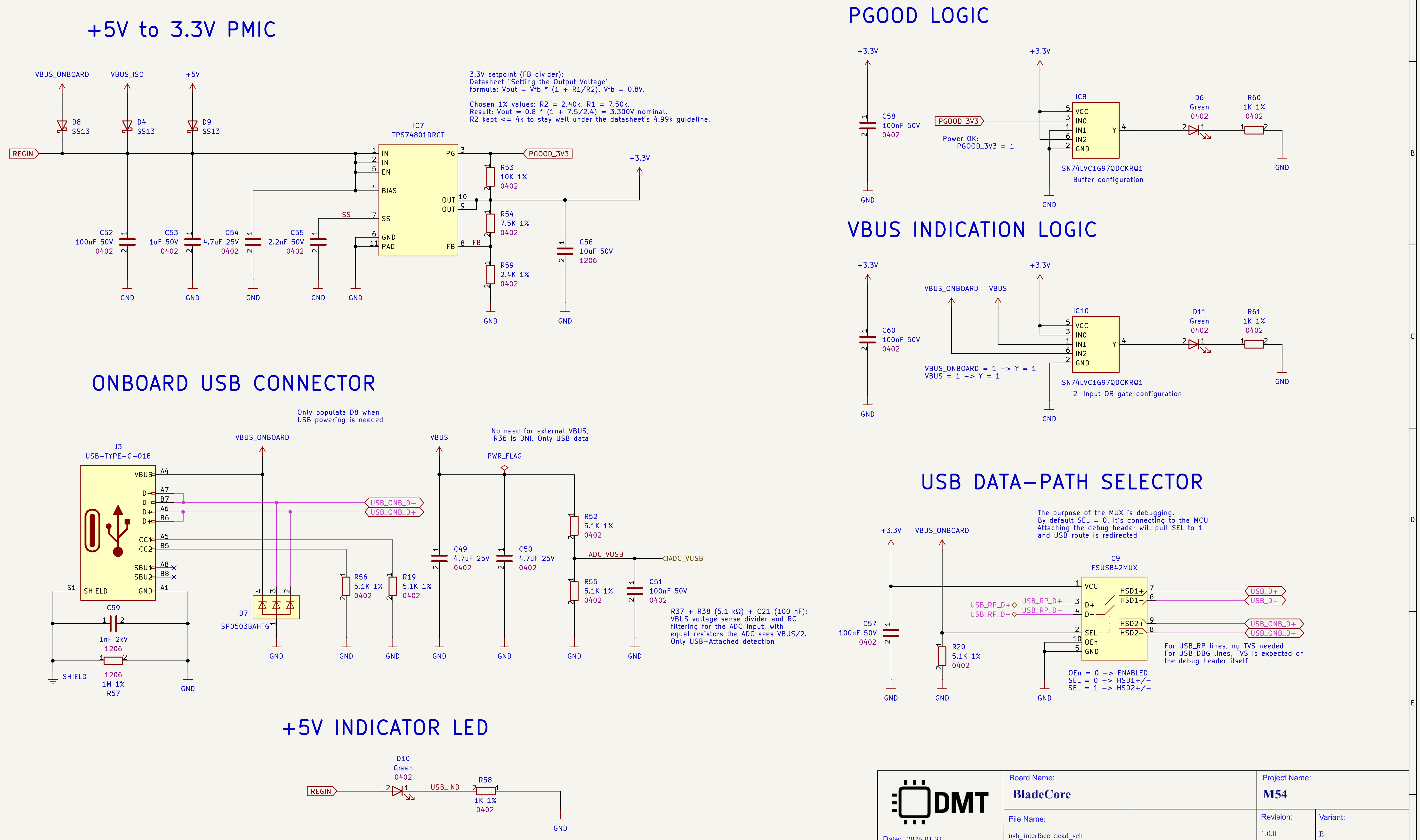


DECOUPLING



DMT	Board Name: BladeCore		Project Name: M54	
Date: 2026-01-31	File Name: ethernet_interface.kicad_sch		Revision: 1.0.0	Variant: E
Sheet Title: Ethernet Interface	Company: DvidMakesThings	Designer: David Sipos Reviewer:	Size: A3	Sheet: 6 of 9

[7] USB Interface and Power



DMT	Board Name: BladeCore		Project Name: M54
Date: 2026-01-31	File Name: usb_interface.kicad_sch	Revision: 1.0.0	Variant: E
Sheet Title: USB Interface and Power	Company: DvidMakesThings	Designer: David Sipos Reviewer:	Size: A3
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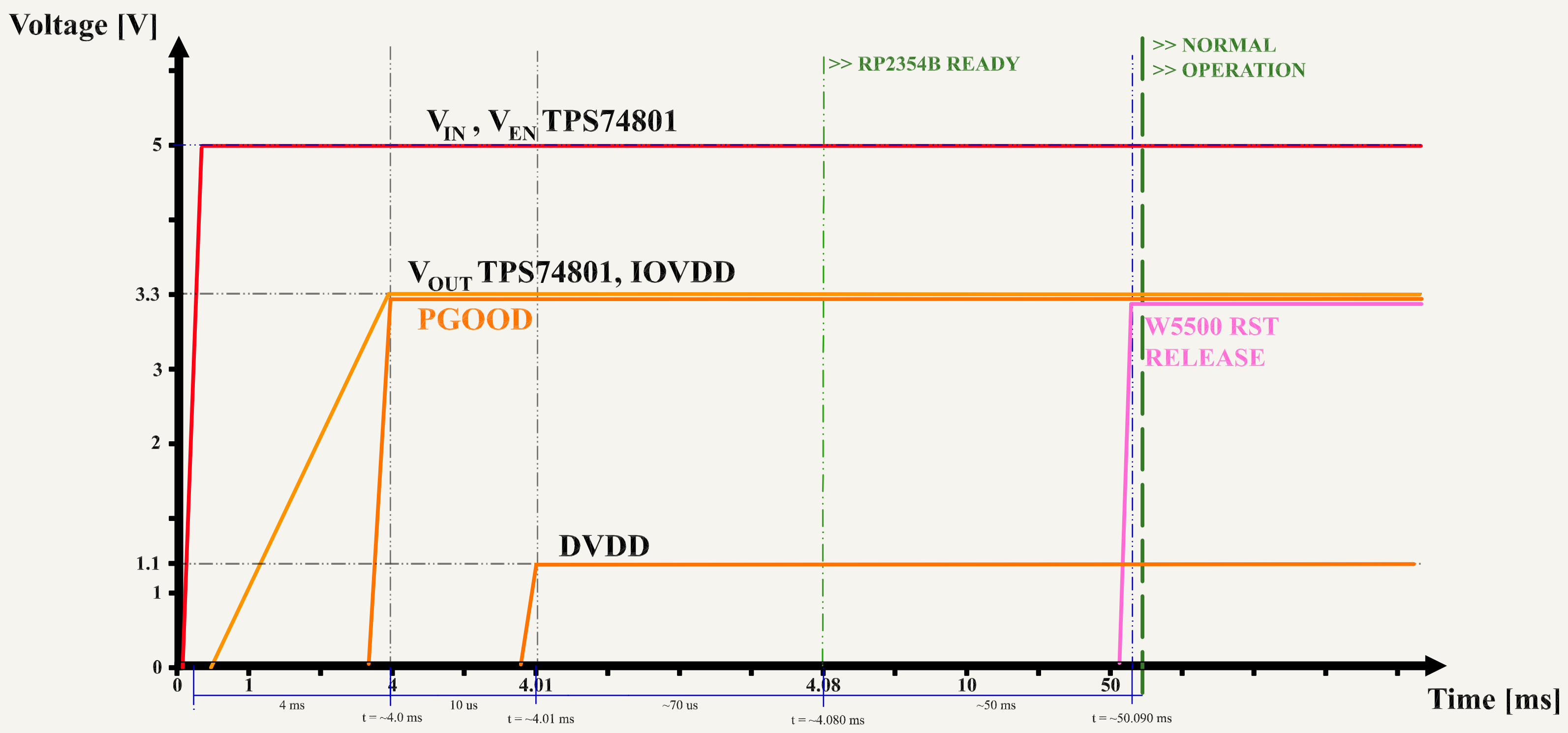
[8] Power - Sequencing



#	Stage	Δt	Total elapsed
1	VIN applied (5.0V)	0	0.000 ms
2	VEN high (TPS74801 EN)	0	0.000 ms
3	VOUT = 3.3V (soft-start)	4.000 ms	4.000 ms
4	PGOOD valid	0	4.000 ms
5	RP2354B 3.3V present	0	4.000 ms
6	RP2354B 1.1V valid (POR)	0.010 ms	4.010 ms
7	RP2354B ready	0.080 ms	4.090 ms
8	W5500 ready	50.000 ms	54.090 ms

DMT	Board Name: BladeCore		Project Name: M54	
Date: 2026-01-31	File Name: Power - Sequencing.kicad_sch		Revision: 1.0.0	Variant: E
Sheet Title: Power - Sequencing	Company: DvidMakesThings	Designer: David Sipos Reviewer:	Size: A3	Sheet: 8 of 9

[9] Revision History

DATE	REVISION	RESPONSIBLE	CHANGE
31.01.2026	1.0.0	DMT	INITIAL CREATION

DMT

Date: 2026-01-31

Board Name: BladeCore

Project Name: M54

File Name: Revision History.kicad_sch

Revision: 1.0.0

Variant: E

Sheet Title: Revision History

Company: DvidMakesThings

Designer: David Sipos

Reviewer:

Size: A3

Sheet: 9 of 9